

System Analyzer

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Explanation

My script scans a computer for file names that contain key words. The script crawls through all directories on a users computer starting from their home directory. As different files are being approached the script checks if that file name contains certain keywords that are in a list of keywords. It also creates a file called information report that is a list of potential files that contain keywords. These files are then added to a zip file called information.zip.

CODE SNIPPETS

CodeFileEditSelectionViewGoRunTerminalWindowHelp

CS 3080

EXPLORER

OPEN EDITORS

CS 3080

assignment1.py.zip

btc_order_book_summary.csv

Categorical_test_results.csv

Categorical_train_results.csv

class.py

CosineSimilarity.py

dorking_script.py

fetch.py

httpServer.py

information.zip

Linear_test_results.csv

Linear_train_results.csv

llm.py

Logistic_test_results.csv

Logistic_train_results.csv

Neural_test_results.csv

Neural_train_results.csv

NN_trevino.py

oopPractice.py

part_b_worksheet2.py

practice.py

sensitive_file_report.txt

session8.py

Socket.py

studypractice.py

systemAnalyzer.py

test_energy_data.csv

train_energy_data.csv

Trevino

Trevino_Harvey_session10.py

trevino_session9.py

Trevino_worksheet7.py

wks3.py

wks11.py

OUTLINE

TIMELINE

systemAnalyzer.py > ...

1

2 import os

3 import shutil

4 import zipfile

5 from datetime import datetime

6

7 SENSITIVE_KEYWORDS = [

8 "password", "passwd", "secret", "token", "key",

9 "credential", "login", "api", "private"

10]

11

12 SENSITIVE_EXTENSIONS = [

13 ".env", ".key", ".pem", ".crt", ".pfx",

14 ".sql", ".db", ".sqlite", ".bak", ".config"

15]

16

17

18 def get_size(path):

19 total_size = 0

20

21 if os.path.isfile(path):

22 return os.path.getsize(path)

23

24 for root, dirs, files in os.walk(path):

25 for file in files:

26 file_path = os.path.join(root, file)

27

28 if os.path.exists(file_path) and not os.path.islink(file_path):

29 total_size += os.path.getsize(file_path)

30

31 return total_size

32

33

34 def is_sensitive_file(filename):

PROBLEMS

OUTPUT

DEBUG CONSOLE

TERMINAL

PORTS

Python +

Python 3.10.20

(base) htrevinopatton@Harveys-MacBook-Pro CS 3080 % ls

Categorical_test_results.csv

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experiments1

experiments2

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wks11.py

wks12

wks3.py

worksheet4.py

worksheet5.py

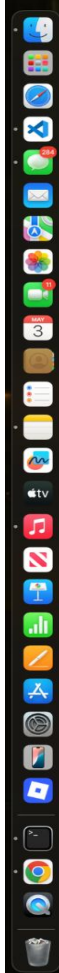
wowfiles.py

trevino_session9.py

Trevino_worksheet7.py

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OUTLINE

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systemAnalyzer.py > ...

60 return [folder for folder in folders if os.path.exists(folder)]

61

62

63 def create_report():

64 report_name = "sensitive_file_report.txt"

65

66 with open(report_name, "w") as report:

67 report.write("User File Security Scan Report\n")

68 report.write(f"Scan Date: {datetime.now()}\n")

69 report.write("=" * 50 + "\n\n")

70

71 for folder in get_user_folders():

72 folder_size = get_size(folder)

73

74 report.write(f"[DIRECTORY] {folder}\n")

75 report.write(f"Size: {folder_size} bytes\n\n")

76

77 for root, dirs, files in os.walk(folder):

78 for file in files:

79 file_path = os.path.join(root, file)

80

81 if os.path.exists(file_path) and not os.path.islink(file_path):

82 file_size = os.path.getsize(file_path)

83

84 if is_sensitive_file(file):

85

86 report.write("[POSSIBLE SENSITIVE FILE]\n")

87 report.write(f"Path: {file_path}\n")

88 report.write(f"Size: {file_size} bytes\n\n")

89

90

91 return report_name

92

93 def copymplease():

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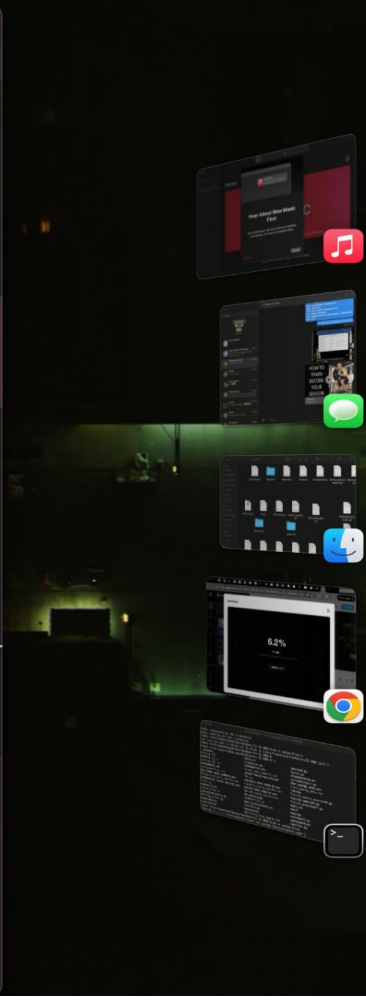
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OUTLINE

TIMELINE

```
108
109 def create_zip_with_files():
110     destination_folder = "copied_files"
111     os.makedirs(destination_folder, exist_ok=True)
112
113     zip_name = "information.zip"
114     home = os.path.expanduser("~")
115
116     with zipfile.ZipFile(zip_name, "w", zipfile.ZIP_DEFLATED) as zipf:
117         for root, dirs, files in os.walk(home):
118             for file in files:
119                 file_path = os.path.join(root, file)
120
121                 if is_sensitive_file(file):
122                     try:
123                         # Keep folder structure to avoid duplicate filenames over
124                         relative_path = os.path.relpath(file_path, home)
125
126                         # Copy file into copied_files folder
127                         dest_path = os.path.join(destination_folder, relative_path)
128                         os.makedirs(os.path.dirname(dest_path), exist_ok=True)
129                         shutil.copy2(file_path, dest_path)
130                         print(f"Copied: {file_path}")
131
132                         # Add copied file to zip
133                         zipf.write(dest_path, arcname=os.path.join("copied_files",
134                             print(f"Added to zip: {relative_path}")
135
136                     except Exception as e:
137                         print(f"Error processing {file_path}: {e}")
138
139     print(f"\nZip created: {zip_name}")
140
141
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

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